

PDBThink

Coordinate Tasks

A verified task pool for Snowball evaluation, RL and teacher-generated SFT

TASK RELEASE v1.3.0

100,000

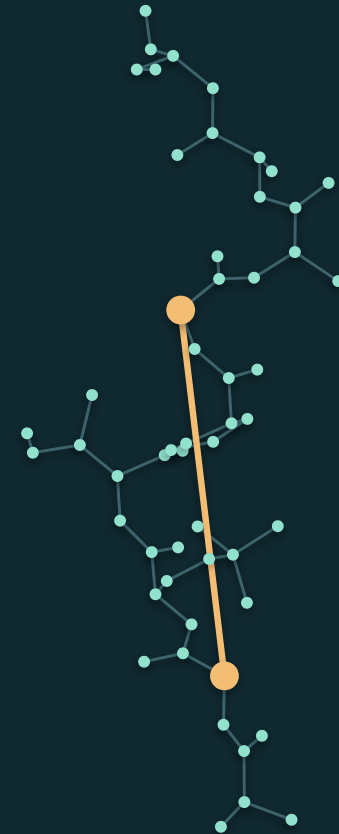
coordinate tasks

19

problem families

28,045

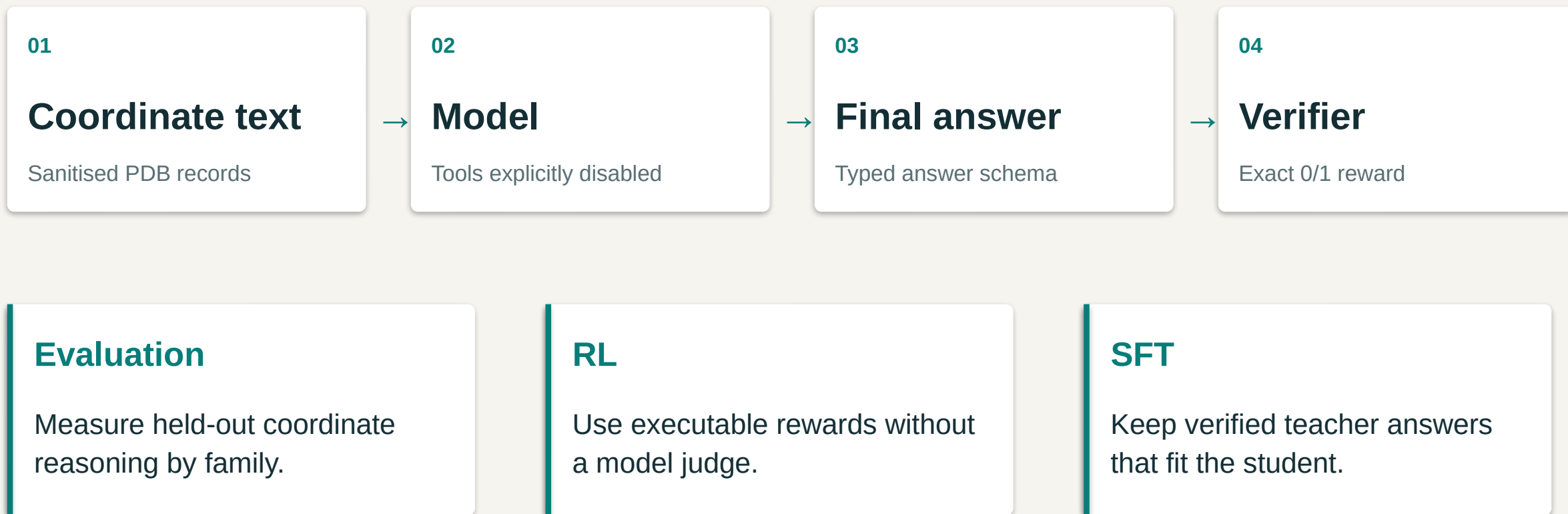
8K-headroom train tasks



Geometry from an actual released task

The answer must follow from the displayed coordinates

One structure (or two states), one question, one tool-free response, one deterministic reward.



Nineteen families span parsing, geometry and structure

All active coordinate families are present. Sequence prediction and retired MECH tasks are excluded.

PARSING

- P01** Chain identifiers
- P02** Residue counting
- P03** Atom coordinates

GEOMETRY

- G01** Atom distance
- G02** Nearest eligible atom
- G03** Nearest residue
- G04** Steric clash

STRUCTURE

- S02** Phosphorylation
- S03** Solvent exposure
- S04** Secondary structure
- S05** Fold class
- S09** Side-chain rotamer

INTERACTIONS

- S01** Salt bridges
- S06** Ligand contacts
- S07** Metal coordination
- S08** Disulfide partner

RELATIONSHIPS

- I01** Interface contacts
- N01** Shared contact
- T01** Two-state contacts

Read two atoms; calculate their distance

G01 · 54 displayed atoms · only the two relevant PDB records are excerpted below.

EXCERPT FROM THE MODEL INPUT

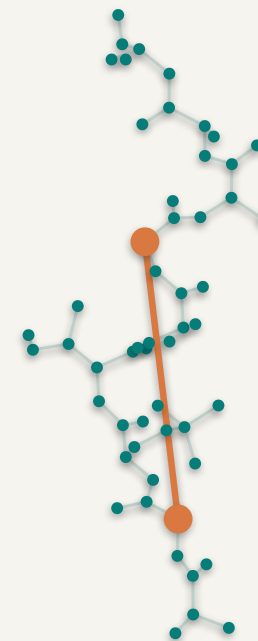
ATOM	7	CA	GLY	P	44	-0.514	7.195	2.794
ATOM	36	CA	GLY	P	48	-0.079	-3.046	-0.922

P:G44:CA $(-0.514, 7.195, 2.794)$

P:G48:CA $(-0.079, -3.046, -0.922)$

$$d = \sqrt{(0.435^2 + 10.241^2 + 3.716^2)} = 10.903 \text{ \AA}$$

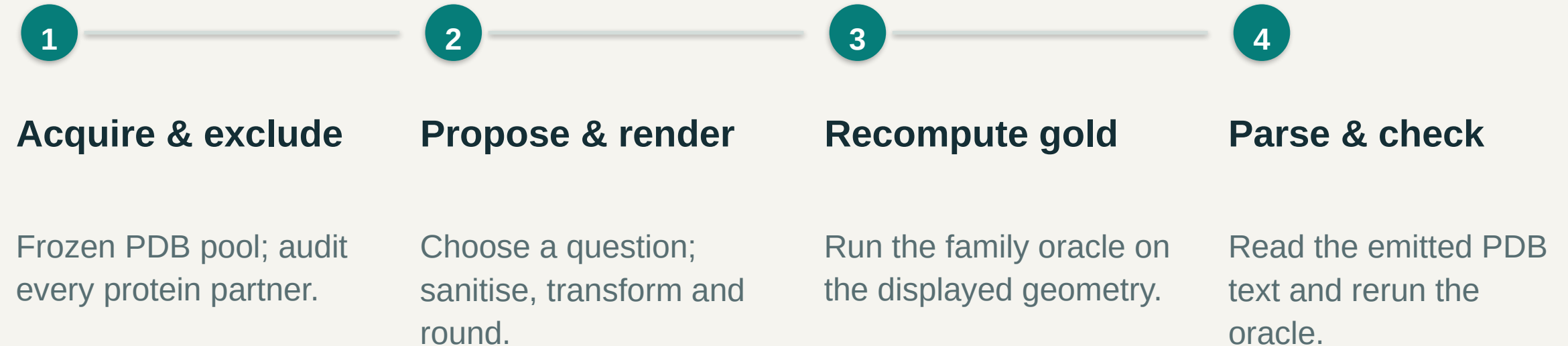
FINAL: 10.90 → reward = 1



The orange line joins the queried atoms.

Check gold against the displayed coordinates

The benchmark's family generators and operational definitions are reused on new source material.



100,000 / 100,000 packaged coordinate checks and oracle-solution checks passed

Hold out related proteins, not random task rows

The release excludes the frozen PDBThink source inventory and keeps related sources in one split.

0

Entry identity

Every configured benchmark source

0

Protein sequence

Full-entity and observed-chain sequences

0

Related-source clusters

RCSB 30% sequence clusters

1,867 source groups

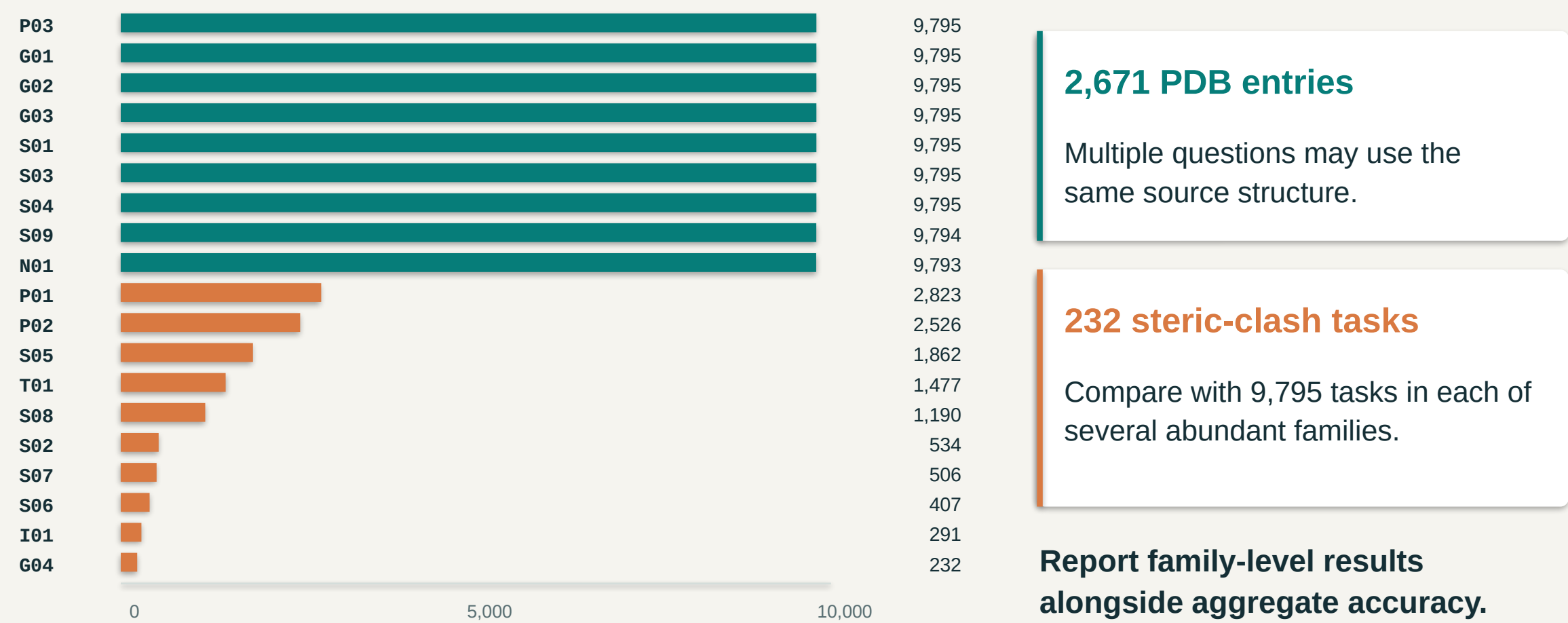
● Train	91,154
● Validation	4,411
● Test	4,435

Audited intersections with the benchmark

This establishes benchmark separation; it does not establish absence from model pretraining.

The 100,000 tasks are deliberately broad, but uneven

Nine abundant families account for 88.2% of tasks. Rare admissible cases limit the remaining families.



Harbor packages a tool-free model task

Task Trove-style Parquet rows contain `task_binary` and a separate `solution_binary`.

MODEL INPUT

prompt.json

System instructions
Exact coordinate question
No gold, provenance or source IDs

One response → FINAL field



HOST / EVALUATOR

Bundled deterministic verifier

Typed parsing + numerical tolerances
Exact correctness → reward 0 or 1
Partial-credit diagnostics kept separately

ORACLE SOLUTION REMAINS SEPARATE

For evaluation, allocate the largest supported output budget that fits the exact native-token prompt.

v1.3 makes the scoring boundary and clash rule explicit

All 100,000 task identities, displayed coordinates, gold answers and grouped splits are preserved.

Inclusive numeric tolerance

−1.710 vs −1.711 should pass ± 0.001 .

Scorer 1.1.0 removes binary-subtraction boundary errors; tolerance is unchanged.

Steric-clash exclusions

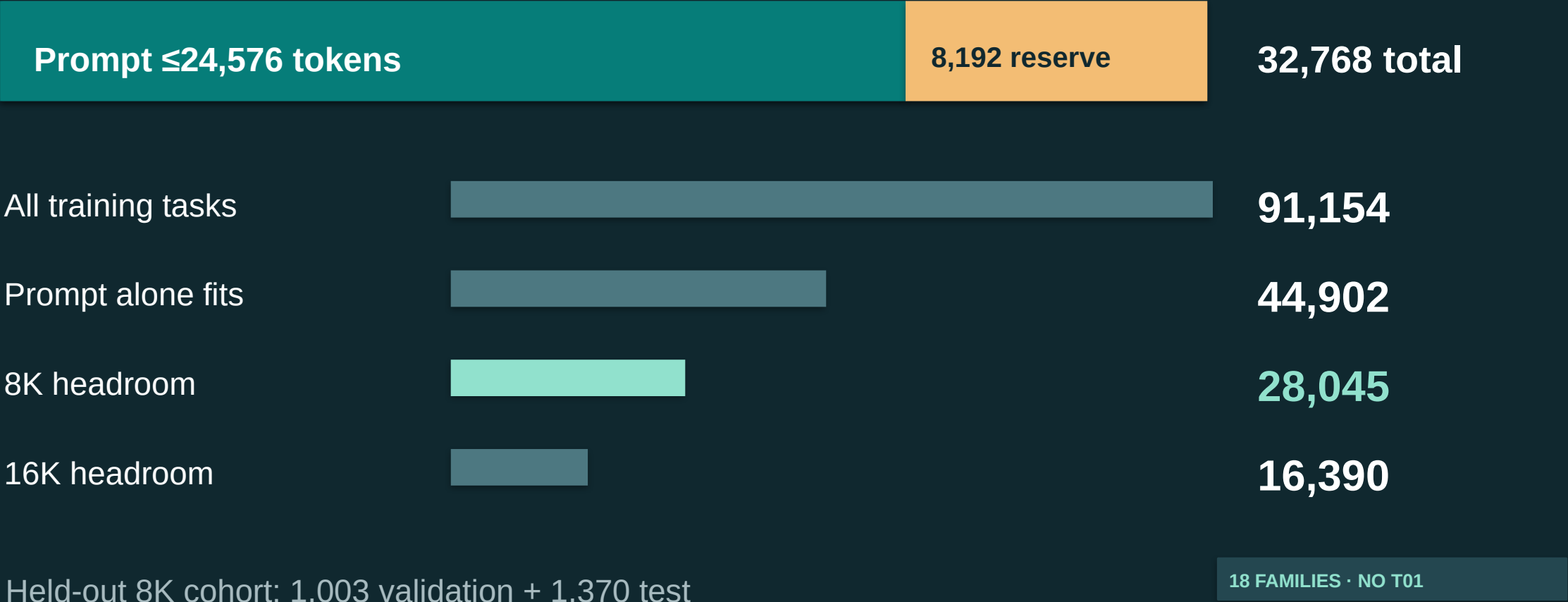
The question now explicitly excludes every SG–SG pair.

Prompt v5 clarifies all 232 G04 tasks without changing their gold answers.

Version boundary Published GLM traces retain their v1.2 prompts and historical rewards.

28,045 training tasks leave 8K for reasoning and output

Exact token counts use Snowball’s pinned tokenizer and thinking-enabled chat template.



GLM samples independently until success or ten attempts

Historical run: GLM-5.3 on the 28,045 eligible training tasks from task release v1.2.0.

Sample

Frozen prompt
Fresh seed
No tools



Score

Task's bundled verifier
Correctness recorded
Raw response retained



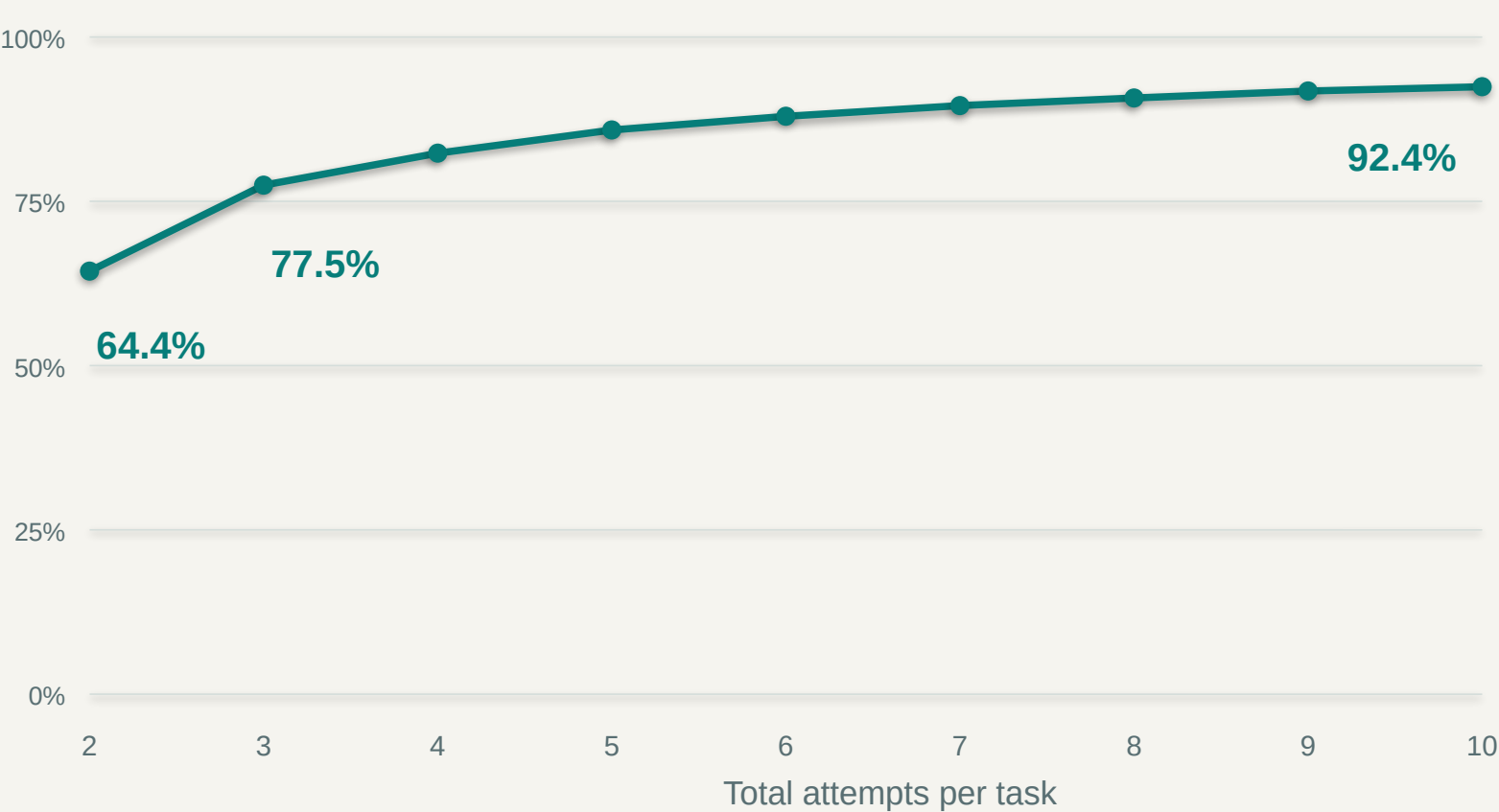
Select or retry

Stop at first success
Otherwise retry unchanged
Maximum ten attempts

8K selects the student cohort; it does not cap the teacher.

Retries recover 1,993 of the 2,156 first-attempt failures

The chart tracks recovery among the 2,156 initially failed tasks · 33,003 total scored attempts.



92.31% → 99.42%

Overall success: first try → within ten attempts.

163 remain unsolved

Ten failed attempts are not proof that a task is impossible.

A correct teacher answer may still be too long for Snowball

The published SFT configuration keeps the first correct response only when the entire conversation fits.



3,360 correct traces exceed context

I01: 0 SFT examples · G04: 12

Use verified SFT first, then test whether RL adds transfer

This is a proposed experiment sequence; no Snowball training gain has been measured here.

01

Reconcile versions

Rescore numeric cases and regenerate G04 on v1.3; publish a new teacher revision.

02

Run a focused SFT pilot

Use verified, student-fitting traces. Compare natural and family-aware mixtures.

03

Add RL on residual errors

Target Snowball's measured weaknesses using deterministic rewards and grouped splits.

Keep the original benchmark and the source-group held-out splits out of training.

Judge progress by held-out transfer and complete answers

Compare baseline Snowball, SFT-only and SFT + RL on the same versioned evaluation cohorts.

Capability

Exact reward by family; macro and task-weighted accuracy.

Transfer

Fixed held-out source groups, plus a separate benchmark readout.

Efficiency

Tokens, truncation, format errors and tool-policy compliance.

Coverage

No held-out 8K tasks for ligand contacts, interfaces or two-state changes.

Decision to make

Does RL improve held-out accuracy beyond SFT at an acceptable token cost?

Family inventory and teacher outcomes

Task counts use v1.3.0. Teacher outcomes retain task-v1.2.0 scoring; the 8K training task identities are unchanged.

Family	All tasks	8K train	First try	By ten	SFT fits
P01 Chain identifiers	2,823	918	100.0%	100.0%	918
P02 Residue counting	2,526	434	99.8%	100.0%	434
P03 Atom coordinates	9,795	3,339	97.4%	99.9%	3,336
G01 Atom distance	9,795	3,015	94.9%	99.7%	2,999
G02 Nearest eligible atom	9,795	3,335	94.0%	100.0%	3,247
G03 Nearest residue	9,795	2,892	93.9%	99.9%	2,846
G04 Steric clash	232	58	51.7%	87.9%	12
S01 Salt bridges	9,795	1,181	99.3%	100.0%	1,180
S02 Phosphorylation	534	436	89.9%	100.0%	436
S03 Solvent exposure	9,795	3,334	87.5%	97.6%	2,307
S04 Secondary structure	9,795	3,124	82.7%	98.5%	1,954
S05 Fold class	1,862	314	84.7%	99.7%	82
S06 Ligand contacts	407	53	45.3%	100.0%	33
S07 Metal coordination	506	92	45.7%	90.2%	81
S08 Disulfide partner	1,190	278	98.6%	100.0%	278
S09 Side-chain rotamer	9,794	3,285	90.4%	100.0%	3,117
I01 Interface contacts	291	12	33.3%	66.7%	0
N01 Shared contact	9,793	1,945	97.6%	100.0%	1,262
T01 Two-state contacts	1,477	0	—	—	0

Load pinned releases, then enforce the scientific contract

Keep task and teacher versions explicit in every training and evaluation manifest.

```
from datasets import load_dataset

tasks = load_dataset(
    "open-athena/pdbthink-coordinate-tasks",
    split="train", revision="3734406cb97b1702844319f9a5d860cbbf8fe660"
)

traces = load_dataset(
    "open-athena/pdbthink-glm53-teacher-traces", "sft",
    split="train", revision="91baf80abe4cd10e3258bbcc0addafd4d3d32b22"
)
```

For evaluation / RL

Read prompt.json; run the bundled verifier outside the model.

For SFT

Use messages, assistant-only loss and exact token-length checks.

Trace configurations: sft = selected targets · attempts = all scored responses · outcomes = all cohort tasks

Pinned evidence and version boundaries

The deck is reproducible from evidence.json; full references are also in the speaker notes.

TASKS

PDBThink Coordinate Tasks · v1.3.0

3734406cb97b1702844319f9a5d860cbbf8fe660

100,000 tasks · prompt v5 · scorer 1.1.0 · definitions v1.0.0

TEACHER

GLM-5.3 Teacher Traces · v1.0.0

91baf80abe4cd10e3258bbcc0addafd4d3d32b22

Historical prompts and rewards from coordinate tasks v1.2.0

SNOWBALL

Snowball-67B-A2B-5.7T-Mixed-RLVR-Step38

cfc1d845dae89b067cdc7250d0164abefa5a69cf

Native tokenizer and thinking-enabled template · 32,768-token window

More evidence: [manifest](#) · [native-context audit](#) · [correction audit](#) · [teacher summary](#) · [frozen generator code](#)

[Links are clickable in the deck and PDF.](#)